Introduction to Distributed Systems

A distributed system is a collection of independent computers that appears to its users as a single coherent system. The key goals are transparency, scalability, and reliability. Consensus protocols allow nodes to agree on a single value even when some nodes may fail.

Replication and Consistency

Replication improves availability but introduces the challenge of keeping replicas consistent. Eventual consistency allows temporary divergence between copies. Strong consistency guarantees that all reads return the latest committed write at the cost of latency.